

Module 0: Mathematical Concepts in ML/AI

Quick Reference Guide

Learning Outcomes:

1. Apply linear algebra techniques to represent and manipulate data in ML contexts.
2. Apply key calculus principles to support model development and optimization in ML/AI.
3. Evaluate core optimization strategies used to improve ML model performance.
4. Implement basic optimization workflows in Python to solve ML problems

Linear algebra

- **Vectors**
 - Represent features, predictions and model parameters
 - Operations
 - Addition: $a + b = (a_1 + b_1, \dots, a_n + b_n)$
 - Scalar multiplication: $k \cdot v = (kv_1, \dots, kv_2)$
 - Dot product: $a \cdot b = \sum a_i b_i$
- **Matrices**
 - Used for representing data and transformations
 - Matrix multiplication: rows of A and columns of B
 - Identity matrix: I
 - Zero matrix: 0
- **Eigenvalue and eigenvectors**
 - $Av = \lambda v$
 - Used in PCA, data compression and stability analysis
- **Matrix decomposition**
 - SVD: $Av = U \Sigma V^T$
 - LU: $A = LU$
 - Enables efficient computation and simplification

Calculus

- **Derivatives**
 - Measure the rate of change
 - $f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right]$
- **Partial derivatives**

- Derivatives with respect to one variable at a time
- $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$
- **Chain rule**
 - Composite functions: $h(x) = f(g(x)) \rightarrow h'(x) = f'(g(x)) \cdot g'(x)$
- **Norms**
 - Measure vector length or distance
 - $L_1: \|x\|_1 = \sum |x_i|$
 - $L_2: \|x\|_2 = \sqrt{\sum x_i^2}$
 - $L_\infty: \|x\|_\infty = \max(|x_i|)$

Probability and statistics

- **Distributions**
 - Model randomness (e.g. binomial, normal)
- **Expectation and variance**
 - $E[X] = \sum x \cdot P(x)$
 - $Var(X) = E[(X - E[X])^2]$
- **Entropy**
 - Measures uncertainty
 - $H(X) = -\sum p(x) \cdot \log_2 p(x)$

Optimization

- **Gradient descent**
 - Minimises loss
 - $\theta = \theta - \alpha \cdot \nabla J(\theta)$, in which α = learning rate
- **Loss functions**
 - MSE: $\frac{1}{n} \sum (y_i - \hat{y}_i)^2$
 - Cross-entropy: $-\sum y \cdot \log \hat{y}$
- **Activation functions**
 - Sigmoid: $\frac{1}{(1+e^{-x})}$
 - ReLU: $\max(0, x)$
- **Regularization**
 - Prevents overfitting

- L1: adds $\lambda \sum |w_i|$
- L2: adds $\lambda \sum w_i^2$